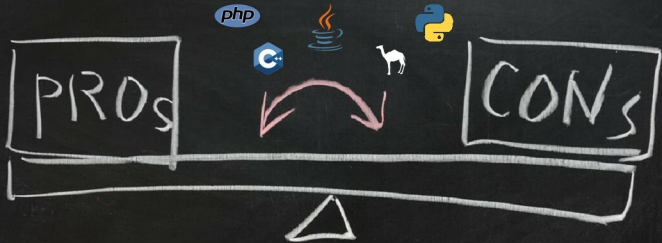


The Pros and Cons of Open Source Programming Languages



You can always look forward to a good debate when the topic is 'Open source programming languages versus licensed programming languages'. There are pros and cons to both sides of the argument. The author gives readers, particularly aspirants in the field of programming, insights into this age old question.

In the current tech savvy scenario, we have become too busy to spend time on manual processes and hence try to get the system to perform tasks to the greatest extent possible. This means we must convey the instructions to the system, using various languages. The programming language plays an important role in letting us manipulate the system or the machine. There are different programming languages currently in use, of which some are open source while others are proprietary.

Open source basically refers to a program in which the source code is available to the public, free of cost, for use or for modification from its original design. It's actually developed by a combined effort, under which programmers improve upon the code and share the changes with the public. As per the Open Source Initiative (OSI), the source code of an open source computer program is made available free of cost to the public so that the larger group of programmers who are not concerned about proprietary rights or financial gains will produce a more useful product that can be utilised further by everyone.

Eligibility criteria

According to DFSG (Debian Free Software Guidelines), an open source programming language should fulfil the following criteria:

1. **Free redistribution:** An open source programming language should not restrict anyone from selling or

giving away any component of it from an aggregate distribution containing various components from several different sources. It should also not require any fee for such sale.

2. **Source code:** An open source programming language must include the source code. If, in case, some form of a product is not distributed with the source code, then there must be some means of obtaining the source code for a reasonable reproduction cost, like downloading via the Internet without any charge. The source code must be the preferred form in which a programmer can modify the program. Obfuscated source code is not allowed.
3. **Derived works:** An open source programming language must be used to make any changes or derived works, and also, those changes must be permitted to be distributed under the same terms and conditions as that of the original programming language.
4. **Integrity of the author's source code:** An open source programming language may restrict its source code from being distributed in modified form only if it allows the distribution of its patch files with its source code for the purpose of modifying the program developed during build time. It must explicitly permit its distribution built from modified source code. It may also require derived works to carry a different name or version from the original software.